Iris Flower Classification using SVM

1. Introduction

Support Vector Machine (SVM) is a supervised machine learning algorithm widely used for classification tasks. It works by finding the optimal hyperplane that separates data points of different classes with the maximum margin. The Iris dataset, a classic dataset in machine learning, is used here to demonstrate SVM's effectiveness in classifying flower species.

2. Working Principle

- **Hyperplane**

A hyperplane is a decision boundary that separates different classes.

- **Margin**

Distance between hyperplane and nearest data points

SVM maximizes this margin

- **Support Vectors**

Closest data points to the hyperplane

These points determine the position of the boundary

- **Kernel Trick**

Used when data is not linearly separable:

Linear Kernel

Polynomial Kernel

RBF Kernel

3. Dataset

The Iris dataset contains:

- **150 samples**

4 features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- **3 classes:**
- Setosa
- Versicolor
- Virginica

4. Steps

Step 1: Import Libraries

- NumPy
- Pandas
- Scikit-learn
- Joblib

Step 2: Load Dataset

Dataset is loaded using sklearn:

```
from sklearn.datasets import load_iris
iris = load_iris()
```

Step 3: Data Preprocessing

- Features (X) → input variables
- Target (y) → output labels

Step 4: Train-Test Split

Dataset is divided into:

- Training set
- Testing set

Step 5: Model Training

SVM model is created using:

```
from sklearn.svm import SVC
model = SVC(kernel='linear')
model.fit(X_train, y_train)
```

Step 6: Prediction

Model predicts class labels for test data.

Step 7: Model Evaluation

Performance is evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

Step 8: Model Saving

Model is saved using joblib:

```
joblib.dump(model, "svm_model.pkl")
```

5. Results

- **Accuracy**: ~0.97 (depending on random split).
- **Confusion Matrix**: Shows correct classification of most samples with minimal misclassifications.
- **Classification Report**: High precision, recall, and F1-score across all three Iris species.

6. Streamlit Application

The application includes:

❖ **Dataset Overview**

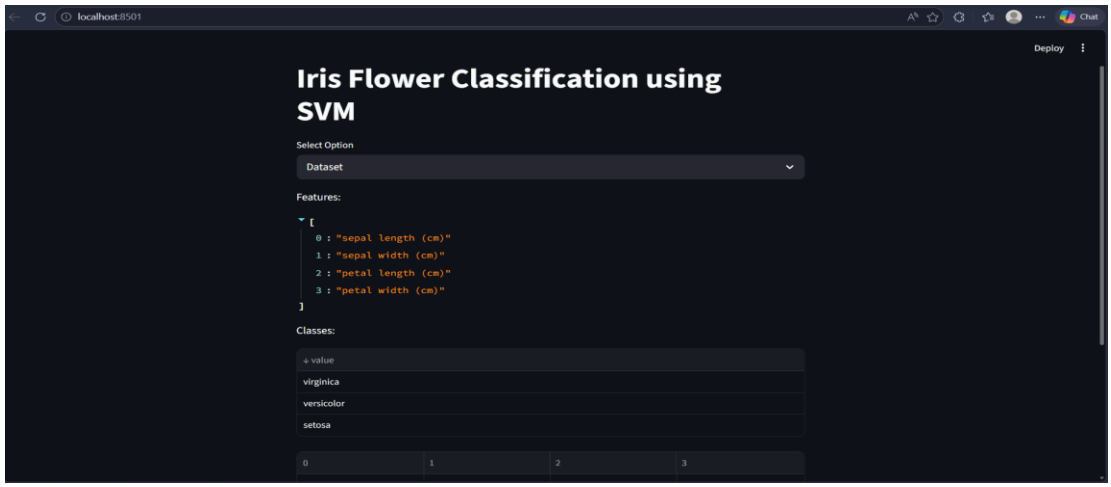
- Displays dataset samples
- Shows features and classes

❖ **Model Explanation**

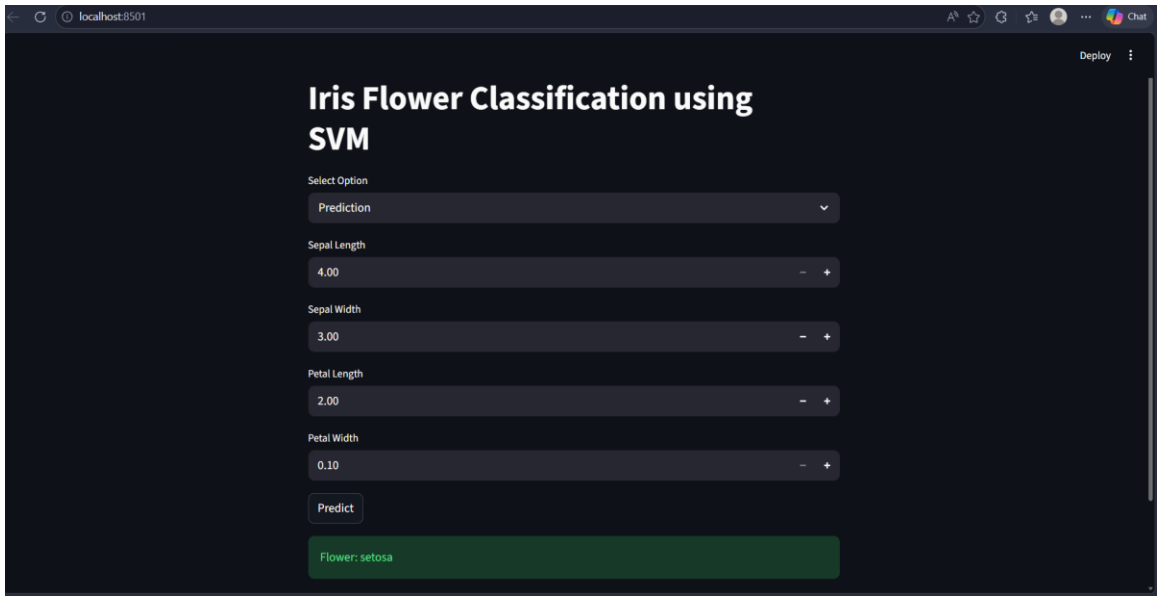
- Provides description of SVM algorithm

❖ **Prediction Section**

- User inputs feature values
- Model predicts flower type


localhost:8501
Chat
Deploy
Iris Flower Classification using SVM
Select Option
Dataset
Features:
[
0 : "sepal length (cm)"
1 : "sepal width (cm)"
2 : "petal length (cm)"
3 : "petal width (cm)"
]
Classes:
value
virginica
versicolor
setosa
0
1
2
3

localhost:8501
Chat
Deploy
Iris Flower Classification using SVM
Select Option
Model
SVM is used for classification.
Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification. It finds the best boundary (hyperplane) that separates different classes with maximum margin.


localhost:8501
Chat
Deploy
Iris Flower Classification using SVM
Select Option
Prediction
Sepal Length
4.00
Sepal Width
3.00
Petal Length
2.00
Petal Width
0.10
Predict
Flower: setosa

7. Conclusion

SVM is an effective algorithm for classification problems. In this project, it provided high accuracy and was successfully integrated into a user-friendly Streamlit application.